Mobile App Developer – Take-Home Assignment
Focus: Architecture, Code Quality, AI-Assisted Development, Ownership
Time Expectation: 4–6 hours (maximum)
Submission Window: 48–72 hours
Platform: Android / iOS / Flutter / React Native (choose one you are most comfortable with)

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1. Objective
This assignment is designed to evaluate how you:

- Solve real-world problems
- Design clean and scalable mobile architecture
- Use AI tools thoughtfully and responsibly
- Make and explain engineering trade-offs
- Communicate decisions through documentation

This is **not** a speed test or a UI-heavy challenge.

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2. Problem Statement
Build a **simple, production-ready mobile application** that displays a list of items fetched from a **public REST API**, with **offline support** and basic user interactions.
Example Use Cases (choose one)

- Task list
- Product catalog
- News articles
- Orders / items list

You may choose **any public REST API**, but it must be clearly documented.

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3. Functional Requirements (Mandatory)
Core Features

- Fetch data from a public REST API
- Display a **list screen**
- Implement **pagination or lazy loading**
- Display a **detail screen** for a selected item
- **Offline support** (cache the last successful response)
- Handle:
  - Loading state
  - Empty state
  - Error state (network / API failure)

Platform & Architecture

- Use a **modern architecture** (MVVM, MVI, Clean, etc.)
- Clear separation of concerns
- Maintainable and scalable structure

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4. Non-Functional Requirements

- Clean, readable, maintainable code
- Modular structure
- Meaningful naming
- Production-level error handling
- Comments where intent is not obvious
- No hardcoded secrets or credentials

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5. AI Usage Policy (Important)

✅ **Allowed AI Usage**

You are encouraged to use AI tools for:

- Boilerplate generation
- Architecture brainstorming
- Refactoring
- Debugging
- Writing unit tests
- Documentation drafting

Examples:

- ChatGPT
- GitHub Copilot
- Cursor
- Claude

❌ **Disallowed AI Usage**

- Blind copy-pasting without understanding
- Generating the entire solution without modification
- Submitting code you cannot explain
- Using paid templates or full starter apps without disclosure

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6. Mandatory AI Disclosure

You **must** include a file named:

AI_USAGE.md

Include:

- AI tools used
- Where AI was used (files / modules)
- What AI output was accepted vs modified
- What AI suggestions were rejected and why
- One example where you improved AI output using your own judgment

⚠️ **Missing this file will result in rejection**

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7. Required Deliverables

1. **Source code repository** (GitHub / GitLab)
2. **README.md** containing:
    - Setup instructions
    - Architecture overview
    - Key decisions and trade-offs
    - Known limitations
3. **AI_USAGE.md** (mandatory)
4. Screenshots or a short screen recording of the app

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8. Optional (Bonus – Not Required)

- Unit tests
- Simple UI animations
- Linting or formatting setup
- Basic CI/CD configuration

Skipping bonus items will **not** negatively impact your evaluation.

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9. What We'll Discuss in the Next Round

- Architecture and design decisions
- AI-generated sections of the code
- Trade-offs you made
- What you would improve with more time
- How you validated and refined AI output

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10. Evaluation Focus

We will evaluate:

- Architecture and structure
- Code quality and readability
- Thoughtful AI usage
- Problem-solving and edge-case handling
- Documentation clarity
- UX basics (loading, error, empty states)

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11. Submission Instructions

- Share the repository link via email kiara.dave@lumoslogic.com
- Ensure all required files are present
- Ensure the project builds and runs successfully
- Candidates must upload the code to github and share the link in the email. Candidates must submit this task in 2 days.

We value clarity, ownership, and judgment over completeness or polish.

Good luck, and we look forward to reviewing your submission.